

QA DISPOSITION REPORT burn-in parametric screening

Component	LOT233-C0222	Lot	LOT233	Board	LOT233-B03
Socket	r3 c6	Model version	dataset-v1.1 / ModuleA-d13e1ef / ModuleB-2a28c092619:01		

DECISION: WATCH

LAYER 1 — deterministic evidence (every value cites its comparison basis)

Absolute limit	PASS (8.73 ns vs 10.0 ns max, prop_delay_ns)
Lot-relative level	+0.9 robust sigma vs lot median on leakage_na (median 26.46 nA, sigma 5.59 nA, n=499, MAD estimator; AEC-Q001 IQR/1.35 gives 5.70 nA)
Drift 0h to 24h	+1.483 mV on vth_shift_mv; lot median drift +1.001 mV, part is +0.6 robust sigma on the lot's drift distribution
Drift rate	0.0618 mV/h (lot median 0.0417 mV/h, 1.5x)
Max interval jump	+0.7 robust sigma on iddq_ua; largest interval change is 3.5x the median of the others (concentrated -> step)
Predicted 168h	31.538 nA (leakage_na, from 0h and 24h only)
95% upper bound	33.728 nA -> within lot-derived safe limit 63.645 nA
Joint anomaly	Mahalanobis D2 = 23.0 (good-part 99th pct = 19.7); largest contribution leakage_na at 36% of the total
Leakage/delay sign	leakage +0.9 sigma, delay +0.7 sigma -- consistent with the lot's process locus
Spatial check	no neighbour clustering on board LOT233-B03 (2/64 flagged), component-specific
Data quality	all checkpoints valid
Primary reason	joint-distribution anomaly across parameters (univariately normal)
Action	Release, but tag the component ID for trend monitoring at the next screen. No rework.

LAYER 2 — model attribution

detector	normalised strength	fired	feature (Module B)	SHAP (mV)
L3a_MCD	1.34	YES	v24	-2.285
L3b_Q	0.93	no	z:d1	-0.119
L4d_AutoEnc	0.19	no	s1	+0.116
L4b_LOF	-0.02	no	d1	+0.082
L1_dynamic_MAD	-0.15	no	lotmed_t0	-0.077

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

This part would have moved from WATCH to PASS if iddq_ua at 96 h had been at or above 11.499 uA (measured 11.373 uA, a change of 0.126 uA = 0.0 robust sigma).

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

localised defect not yet affecting timing paths; the class that escapes univariate screening [confidence: speculative]
Because: every parameter is within its normal range, but the joint distance is D^2=23.0 against a good-part 99th percentile of 19.7
Caveat: This is the weakest mapping in the table and should be labelled as such. A joint-distribution outlier says the part does not resemble its peers; it does NOT say why. The value is in flagging the part for attention, not in naming the mechanism.
Suggested next test: quarantine and prioritise for FA; treat the mechanism as unknown

SIGN-OFF

Inspector		Date	
Disposition		Signature	
Notes			

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.